Ordinary Differential Equations

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1 Integral Curves

An **ordinary differential equation (ODE)** describes a relationship between an unknown function and one or more of its derivatives.

For example,

$$\frac{dy}{dt} = ay - b,$$

where a and b are constants.

A **solution** to a differential equation is a function that satisfies the equation. The graph of a solution is called an **integral curve**.

1.1 A Population Model

Consider the population model

$$\frac{dp}{dt} = rp - k,$$

where $p(t)$ represents the population at time t, r is a proportional growth rate, and k is a constant rate of population loss.

Suppose

$$r = \frac{1}{2}, \qquad k = 450.$$

Then

$$\frac{dp}{dt} = \frac{1}{2}p - 450.$$

Factoring,

$$\frac{dp}{dt} = \frac{1}{2}(p - 900).$$

1.2 Separation of Variables

To solve the equation, separate the variables:

$$\frac{dp}{p - 900} = \frac{1}{2}\,dt.$$

Integrating both sides,

$$\int \frac{dp}{p-900} = \int \frac{1}{2}\, dt.$$

Therefore,

$$\ln|p-900| = \frac{t}{2} + C.$$

Exponentiating,

$$|p-900| = e^{t/2+C}.$$

Since the constant can absorb the sign, we write

$$p - 900 = \alpha e^{t/2},$$

where α is an arbitrary constant.

Thus,

$$\boxed{p(t) = 900 + \alpha e^{t/2}}.$$

1.3 Applying an Initial Condition

Suppose the initial population is

$$p(0) = p_0.$$

Substituting $t = 0$,

$$p_0 = 900 + \alpha e^0.$$

Since $e^0 = 1$,

$$p_0 = 900 + \alpha.$$

Therefore,

$$\alpha = p_0 - 900.$$

The solution satisfying the initial condition is

$$\boxed{p(t) = 900 + (p_0 - 900)e^{t/2}}.$$

1.4 Equilibrium Population

An equilibrium occurs when the population does not change. Thus,

$$\frac{dp}{dt} = 0.$$

For the population model,

$$\frac{1}{2}p - 450 = 0.$$

Solving,

$$p = 900.$$

Therefore,

$$\boxed{p = 900}$$

is the equilibrium population.

Notice that the general solution

$$p(t) = 900 + \alpha e^{t/2}$$

contains the equilibrium solution when $\alpha = 0$.

2 The General Equation $dy/dt = ay - b$

Consider the differential equation

$$\frac{dy}{dt} = ay - b,$$

where a and b are constants.

Factor out a:

$$\frac{dy}{dt} = a\left(y - \frac{b}{a}\right).$$

Assuming $a \neq 0$, separate the variables:

$$\frac{dy}{y - b/a} = a\,dt.$$

Integrating both sides,

$$\int \frac{dy}{y - b/a} = \int a\, dt.$$

Therefore,

$$\ln \left| y - \frac{b}{a} \right| = at + C.$$

Exponentiating,

$$y - \frac{b}{a} = \alpha e^{at}.$$

Thus,

$$\boxed{y(t) = \frac{b}{a} + \alpha e^{at}}.$$

2.1 Applying an Initial Condition

Suppose

$$y(0) = y_0.$$

Then

$$y_0 = \frac{b}{a} + \alpha e^0.$$

Since $e^0 = 1$,

$$y_0 = \frac{b}{a} + \alpha.$$

Therefore,

$$\alpha = y_0 - \frac{b}{a}.$$

Hence,

$$\boxed{y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a} \right) e^{at}}.$$

3 Differential Equations and Initial-Value Problems

It is important to distinguish between a differential equation and an initial-value problem.

A **differential equation** specifies a relationship involving an unknown function and its derivatives.

For example,

$$\frac{dy}{dt} = ay - b.$$

The general solution is

$$y(t) = \frac{b}{a} + \alpha e^{at}.$$

This represents a family of solutions, with each value of α giving a different solution.

An **initial condition** selects one particular solution from this family. For example,

$$y(0) = y_0.$$

Therefore,

$$\boxed{\text{Differential Equation} + \text{Initial Condition} = \text{Initial-Value Problem (IVP)}}.$$

4 Population Growth and Decay

A common population model is

$$\frac{dp}{dt} = rp - k.$$

Here,

- $p(t)$ is the population at time t,
- r is the proportional growth rate,
- k is a constant rate of population loss.

The equilibrium population occurs when

$$\frac{dp}{dt} = 0.$$

Therefore,

$$rp - k = 0,$$

which gives

$$p = \frac{k}{r}.$$

Thus, the equilibrium population is

$$\boxed{p_{\text{eq}} = \frac{k}{r}}.$$

The general solution is

$$\boxed{p(t) = \frac{k}{r} + \left(p_0 - \frac{k}{r}\right) e^{rt}}.$$

4.1 Interpretation of the Solution

The solution

$$p(t) = \frac{k}{r} + \left(p_0 - \frac{k}{r}\right) e^{rt}$$

can be understood by comparing the initial population p_0 with the equilibrium population k/r.

If

$$p_0 = \frac{k}{r},$$

then

$$p(t) = \frac{k}{r}$$

for all t.

If

$$p_0 > \frac{k}{r},$$

then the initial population lies above equilibrium.

If

$$p_0 < \frac{k}{r},$$

then the initial population lies below equilibrium.

5 Falling Objects

A falling object provides another important application of first-order differential equations.

Suppose an object has mass m and is subject to gravity and air resistance. Assume that the drag force is proportional to velocity.

The model is

$$m\frac{dv}{dt} = mg - \gamma v,$$

where

- m is the mass of the object,
- g is the acceleration due to gravity,
- $v(t)$ is the velocity,
- $\gamma > 0$ is the drag coefficient.

Dividing by m gives

$$\frac{dv}{dt} = g - \frac{\gamma}{m}v.$$

This has the same basic form as the previous equations.

5.1 Solving the Falling-Object Equation

Rewrite the equation as

$$\frac{dv}{dt} = -\frac{\gamma}{m}\left(v - \frac{mg}{\gamma}\right).$$

Separate the variables:

$$\frac{dv}{v - mg/\gamma} = -\frac{\gamma}{m}\,dt.$$

Integrating,

$$\ln\left|v - \frac{mg}{\gamma}\right| = -\frac{\gamma}{m}t + C.$$

Exponentiating,

$$v - \frac{mg}{\gamma} = \alpha e^{-\gamma t/m}.$$

Therefore,

$$v(t) = \frac{mg}{\gamma} + \alpha e^{-\gamma t/m}.$$

If the initial velocity is

$$v(0) = v_0,$$

then

$$v_0 = \frac{mg}{\gamma} + \alpha.$$

Thus,

$$\alpha = v_0 - \frac{mg}{\gamma}.$$

The solution is therefore

$$\boxed{v(t) = \frac{mg}{\gamma} + \left(v_0 - \frac{mg}{\gamma}\right) e^{-\gamma t/m}}.$$

5.2 Terminal Velocity

The quantity

$$\boxed{v_T = \frac{mg}{\gamma}}$$

is called the **terminal velocity**.

As $t \to \infty$,

$$e^{-\gamma t/m} \longrightarrow 0.$$

Therefore,

$$\lim_{t\to\infty} v(t) = \frac{mg}{\gamma}.$$

Hence,

$$\boxed{v(t) \longrightarrow \frac{mg}{\gamma} \qquad \text{as } t \to \infty.}$$

Thus, the velocity approaches a limiting value rather than increasing without bound.

5.3 Example: Falling Object

Consider a falling object with

$$g = 9.8, \qquad m = 10, \qquad \gamma = 2.$$

Assume the object is initially at rest, so

$$v(0) = 0.$$

The velocity function is

$$v(t) = \frac{mg}{\gamma} + \left(v_0 - \frac{mg}{\gamma}\right) e^{-\gamma t/m}.$$

Substituting the given values,

$$v(t) = \frac{(10)(9.8)}{2} + \left(0 - \frac{(10)(9.8)}{2}\right) e^{-(2/10)t}.$$

Therefore,

$$\boxed{v(t) = 49 - 49e^{-0.2t}}.$$

5.3.1 Finding the Time to Reach the Ground

Suppose the object is dropped from a height of 300 meters.

Let $s(t)$ denote the distance the object has fallen. Since velocity is the derivative of position,

$$\frac{ds}{dt} = v(t).$$

Thus,

$$\frac{ds}{dt} = 49 - 49e^{-0.2t}, \qquad s(0) = 0.$$

Integrating,

$$\int ds = \int \left(49 - 49e^{-0.2t}\right) dt.$$

Therefore,

$$s(t) = 49t + 245e^{-0.2t} + C.$$

Using $s(0) = 0$,

$$0 = 49(0) + 245e^{0} + C.$$

Hence,

$$C = -245.$$

Therefore,

$$\boxed{s(t) = 49t + 245e^{-0.2t} - 245}.$$

The object reaches the ground when

$$s(t^*) = 300.$$

Thus,

$$49t^* + 245e^{-0.2t^*} - 245 = 300.$$

This equation must be solved numerically. The solution is approximately

$$\boxed{t^* \approx 10.51 \text{ seconds.}}$$

6 Classification of Differential Equations

Differential equations can be classified according to several different properties.

6.1 A. Number of Independent Variables

A differential equation is called an **ordinary differential equation (ODE)** if there is only one independent variable.

For example,

$$\frac{dv}{dt} = g - \frac{\gamma}{m}v$$

is an ODE because the only independent variable is t.

A differential equation with more than one independent variable is called a **partial differential equation (PDE)**.

For example, the heat equation can be written as

$$\boxed{\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}}.$$

Here, $u = u(x, t)$ depends on both x and t.

6.2 B. Number of Unknown Functions

A differential equation may involve a single unknown function or multiple unknown functions.

A collection of differential equations involving multiple unknown functions is called a **system of differential equations**.

For example,

$$\frac{dx}{dt} = \alpha x - \beta xy,$$

$$\frac{dy}{dt} = \gamma xy - \theta y.$$

This is a system involving the two unknown functions $x(t)$ and $y(t)$.

6.3 C. Order of a Differential Equation

The **order** of a differential equation is the order of the highest derivative that appears in the equation.

In general, an nth-order ODE can be written as

$$F\left(t, y, y', y'', \ldots, y^{(n)}\right) = 0.$$

An equation can also be written explicitly in terms of its highest derivative:

$$y^{(n)} = f\left(t, y, y', \ldots, y^{(n-1)}\right).$$

For example,

$$y^{(3)} + 2e^t y'' + y' - t^4 = 0$$

is a third-order differential equation because the highest derivative is $y^{(3)}$.

Another example is

$$(y')^2 + ty' + 4y = 0.$$

This is a first-order differential equation because the highest derivative appearing is y'.

Solving this equation for y' gives

$$(y')^2 + ty' + 4y = 0.$$

Using the quadratic formula,

$$\boxed{y' = \frac{-t \pm \sqrt{t^2 - 16y}}{2}}.$$

6.4 D. Linear vs. Nonlinear

An nth-order differential equation

$$F\left(t, y, y', \ldots, y^{(n)}\right) = 0$$

is **linear** if it is linear in

$$y, \quad y', \quad y'', \quad \ldots, \quad y^{(n)}.$$

The most general form of an nth-order linear differential equation is

$$\boxed{a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \cdots + a_{n-1}(t)y' + a_n(t)y = g(t)}.$$

The functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

are coefficients that depend only on the independent variable t.

6.4.1 Example: Linear Equation

Consider

$$\frac{t}{1+t^2}y'' + 3ty = \sin(t).$$

This equation is

- ordinary, because there is one independent variable t;
- second-order, because the highest derivative is y'';
- linear, because y'' and y appear only to the first power and are not multiplied together.

Therefore, this is an

$$\boxed{\text{ordinary, linear, second-order ODE.}}$$

6.4.2 Examples of Nonlinear Equations

Consider

$$yy' = 0.$$

This equation is **nonlinear** because the unknown function y and its derivative y' are multiplied together.

Similarly,

$$y^{(3)} + 2e^t y'' + yy' = t^4$$

is nonlinear because of the product

$$yy'.$$

Therefore,

$$\boxed{y^{(3)} + 2e^t y'' + yy' = t^4}$$

is a third-order nonlinear ODE.